

GUJARAT TECHNOLOGICAL UNIVERSITY

Diploma Engineering – SEMESTER – 6 (NEW) – EXAMINATION – Winter-2025

Subject Code: 4361602

Date: 15-11-2025

Subject Name: Cloud and Data Center Technologies

Time: 02:30 PM TO 05:00 PM

Total Marks: 70

Instructions:

1. Attempt all questions.
2. Make Suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Use of simple calculators and non-programmable scientific calculators are permitted.
5. English version is authentic.

		Marks
Q.1	(a) Define Cloud computing. Explain any two advantages of using cloud computing.	03
પ્રશ્ન.1	(અ) Cloud computing વ્યાખ્યાયિત કરો. Cloud computing ઉપયોગ કરવાની કોઈપણ બે ફાયદા સમજાવો..	૦૩
	(b) Explain Software as a Service (SaaS).	04
	(બ) Software as a Service (SaaS) સમજાવો..	૦૪
	(c) Explain Fog computing in cloud computing.	07
	(ક) Cloud computing માં Fog Computing સમજાવો.	૦૭
	OR	
	(c) Explain DevSecops in cloud computing.	07
	(ક) Cloud computing માં DevSecops સમજાવો..	૦૭
Q.2	(a) Explain characteristics of virtualization.	03
પ્રશ્ન.2	(અ) Virtualization ની characteristics સમજાવો.	૦૩
	(b) Differentiate Full virtualization, Para virtualization, and Partial virtualization.	04
	(બ) Full virtualization, Para virtualization, and Partial virtualization વચ્ચે તફાવત આપો.	૦૪
	(c) Explain Hypervisor with its types.	07
	(ક) Hypervisor ને તેના પ્રકારો સાથે સમજાવો.	૦૭
	OR	
Q.2	(a) State the advantages of using virtualization. Explain any one.	03
પ્રશ્ન.2	(અ) Virtualization ઉપયોગ કરવાના ફાયદાઓની યાદી બનાવો. કોઈપણ એક સમજાવો.	૦૩
	(b) Explain operating system -level virtualization.	04
	(બ) Operating system -level virtualization સમજાવો.	૦૪
	(c) Explain virtualization of Clusters.	07
	(ક) Clusters નું Virtualization સમજાવો.	૦૭
Q. 3	(a) Define Data Center. List types of Data center.	03
પ્રશ્ન.3	(અ) Data Center ને વ્યાખ્યાયિત કરો. Data center ના પ્રકારોની સૂચિ બનાવો.	૦૩

	(b)	State advantages and disadvantage of SDN (Software Defined Networking).	04
	(બ)	Advantages and disadvantage SDN (Software Defined Networking) ની લખો.	૦૪
	(c)	Explain SDN (Software Defined Networking) architecture.	07
	(ક)	SDN (Software Defined Networking) આર્કિટેક્ચર સમજાવો.	૦૭
		OR	
Q. 3	(a)	List data center network topologies and explain any one of them.	03
પ્રશ્ન.3	(અ)	Data center network topologies જું list લખો અને કોઈ પણ એક સમજાવો.	૦૩
	(b)	Differentiate between scalability and elasticity in cloud data center.	04
	(બ)	Scalability અને elasticity વચ્ચે તફાવત આપો.	૦૪
	(c)	Explain Infrastructure as Code (IaC) with its different approaches.	07
	(ક)	Infrastructure as Code (IaC) ને તેની different approaches સાથે સમજાવો.	૦૭
Q. 4	(a)	What is object storage in the cloud?	03
પ્રશ્ન.4	(અ)	Object storage cloud માં શું છે?	૦૩
	(b)	Explain managed cloud storage and unmanaged cloud storage.	04
	(બ)	Managed cloud storage અને unmanaged cloud storage સમજાવો.	૦૪
	(c)	Explain durability in cloud computing.	07
	(ક)	Cloud computing માં durability સમજાવો.	૦૭
		OR	
Q. 4	(a)	Explain block storage in the cloud.	03
પ્રશ્ન.4	(અ)	Block storage cloud માં સમજાવો.	૦૩
	(b)	Explain data scaling with example.	04
	(બ)	Data scaling વિગતવાર સમજાવો.	૦૪
	(c)	Explain types of cloud databases i)SQL ii) NoSQL.	07
	(ક)	Cloud databases માં i)SQL ii) NoSQL સમજાવો.	૦૭
Q.5	(a)	What are the metrics for Service Level Agreements (SLAs).	03
પ્રશ્ન.5	(અ)	Service Level Agreements(SLAs) ની metrics સમજાવો.	૦૩
	(b)	Explain Serverless Computing.	04
	(બ)	Serverless Computing સમજાવો.	૦૪
	(c)	Explain cloud security challenges.	07
	(ક)	Cloud માં security challenges સમજાવો.	૦૭
		OR	
Q.5	(a)	Explain Securing Private and Public Cloud Architecture.	03
પ્રશ્ન.5	(અ)	Private and Public Cloud Architecture secure કરી રીતે કરી શકાય તે સમજાવો.	૦૩
	(b)	Explain Distributed Ledger Technology (DLT) with Cloud Computing.	04
	(બ)	Distributed Ledger Technology (DLT) ની સાથે Cloud Computing માં સમજાવો.	૦૪
	(c)	Explain access control and authentication in cloud computing.	07
	(ક)	Access control and authentication cloud computing માં સમજાવો.	૦૭